

Ten Strategies for Automation

1. Specialization of operations.
2. Combined operations.
3. Simultaneous operations.
4. Integration of operations.
5. Increased flexibility.
6. Improved material handling and storage.
7. On-line inspection.
8. Process control and optimization.
9. Plant operations control.
10. Computer integrated manufacturing (CIM).

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network diagram

(1) nodes representing operations, work elements, or other entities and
(2) arrows connecting the nodes indicating relationships among the nodes.
The network diagram is used to model work flow between operations.
Critical path method (CPM) and program evaluation and review technique
(PERT) diagrams are network diagrams used to schedule the work activities
in a project.

TRADITIONAL INDUSTRIAL ENGINEERING CHARTING AND DIAGRAMMING TECHNIQUES

1. Operation Charts

2. Process Charts

- flow process chart, used to analyze a material or workpiece being processed.
- worker process chart, used to analyze a worker performing a process. (no storage)
- form process chart, used to analyze the processing of paperwork forms.

3. Flow Diagrams

4. Activity Charts

- Right-hand/left-hand activity chart/a workplace activity chart / Two handed process chart
- Worker-machine activity chart -- Gang activity chart
- Worker-multimachine activity chart / Machine Cluster

Principles of motion economy, they provide guidelines for work design in three categories:

- (1) use of the human body in developing the standard method,
- (2) workplace layout, and
- (3) design of the tooling and equipment used in the task.

PRINCIPLES OF MOTION ECONOMY AND WORK DESIGN

Principles Related to the Use of the Human Body

- Both hands should be fully utilized.
- The two hands should begin and end their motions at the same time.
- The motions of the hands and arms should be symmetrical and simultaneous.
- The work should be designed to emphasize the worker's preferred hand.
- The worker's two hands should never be idle at the same time.
- The method should consist of smooth continuous curved motions rather than straight-line motions with abrupt changes in direction.
- Momentum should be used to facilitate the task wherever possible.
- The method should take advantage of gravity instead of opposing it.
- The method should achieve a natural rhythm of the motions involved.
- The lowest classification of hand and arm motions should be used.
- Minimize eye focus and eye travel activities.
- The method should be designed to utilize the worker's feet and legs when Appropriate.

Principles Related to Workplace Arrangement 7

- Tools and materials should be located in fixed positions within the work area.
- Tools and materials should be located close to where they are used.
- Tools and materials should be placed in locations that are consistent with the sequence of work elements in the work cycle.
- Gravity feed bins should be used to deliver small parts and fasteners.
- Gravity drop chutes should be used for completed work units where appropriate.
- Adequate illumination must be provided for the workplace.
- A proper chair should be provided for the worker.

Principles Related to the Design of Tooling and Equipment

- Workholding devices should be designed for the task; a worker's hand Should not be used as a workholder.
- The hands should be relieved of work elements that can be performed by the feet.
- Multiple tooling functions should be combined into one tool whenever possible.
- Multiple operations should be performed simultaneously rather than Sequentially whenever possible.
- Where feasible, an operation should be performed on multiple parts simultaneously.

→ Equipment controls should be designed for operator convenience and error Avoidance.

→ Hand tools and portable powered tools should be designed for operator Comfort and convenience.

→ Manual operations should be mechanized or automated wherever Economically and technologically feasible.